

- a) What are vanishing and exploding gradients? How do they affect neural networks?
Explain in detail.

Vanishing Gradient

Vanishing gradient occurs when gradients become extremely small during backpropagation. As the error is propagated backward through many layers, the weights of earlier layers receive very small updates and learn very slowly.

Effects:

- Slow or no learning in deep neural networks.
- Early layers fail to capture important features.
- Training becomes inefficient and accuracy may decrease.

Exploding Gradient

Exploding gradient occurs when gradients become very large during backpropagation. This causes large weight updates, making the learning process unstable.

Effects:

- Unstable training process.
- Loss function may fluctuate or become very large.
- Model may fail to converge and produce poor results.

Solutions

- Use ReLU activation instead of sigmoid/tanh.
- Apply proper weight initialization techniques.
- Use Batch Normalization.
- Use Gradient Clipping to control large gradients.
- Employ architectures such as LSTM or GRU for deep sequential networks.

Conclusion

Vanishing and exploding gradients are common challenges in deep neural networks. They affect the learning process by making training either too slow or unstable. Proper activation functions, normalization methods, and optimization techniques help overcome these problems.

b. Bank Churn Rate

Git Repository Link: <https://github.com/MyGit-App/ANN-CNN.git>

Code File Name: Backend.ipynb

Streamlit Code: bankchurn.py

Streamlit App Link: <https://bank-churn-rate.streamlit.app/>

c. CCTV Accident Detection

Git Repository Link: <https://github.com/MyGit-App/ANN-CNN.git>

Code File Name: AccidentDetection.ipynb

Streamlit Code: AccidentDetection.py

Streamlit App Link: <https://accident-detect.streamlit.app/>

d. Explain the role of padding and stride in convolutional layers? How will you decide when to use the padding.

Padding

Padding is the process of adding extra pixels (usually zeros) around the border of the input image before applying the convolution operation.

Purpose:

- Preserves spatial dimensions of the input.
- Prevents loss of information at image boundaries.
- Allows deeper networks without rapidly reducing feature map size.

Stride

Stride is the number of pixels by which the convolution filter moves across the input image.

Purpose:

- Controls the size of the output feature map.
- Reduces computational cost when larger strides are used.
- Helps in downsampling the input features.

When to Use Padding?

- Use **padding = "same"** when it is important to preserve the input dimensions and retain edge information.

- Use **padding = "valid"** (no padding) when dimension reduction is acceptable and computational efficiency is preferred.
- In deep CNNs, padding is commonly used to prevent excessive shrinking of feature maps.

Conclusion

Padding helps preserve information and feature map size, while stride controls the movement of filters and the size of the output. The choice of padding depends on whether maintaining input dimensions is important for the task.

e. How does data augmentation help in improving CNN performance?

Data Augmentation

Data augmentation is a technique used to increase the size and diversity of the training dataset by applying transformations such as rotation, flipping, zooming, shifting, cropping, and brightness adjustment to existing images.

Benefits of Data Augmentation

- Increases the amount of training data without collecting new images.
- Reduces overfitting by exposing the model to varied versions of the same image.
- Improves the model's ability to generalize to unseen data.
- Enhances robustness against changes in image orientation, position, and lighting conditions.
- Leads to better accuracy and performance of CNN models.

Conclusion

Data augmentation improves CNN performance by creating diverse training samples, reducing overfitting, and helping the model learn more robust and generalized features from the data.